

delhi: A Declarative Language and Model Checker for Epistemic Reasoning over Plausibility Models

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July 29, 2026

Abstract

We present *delhi*, a declarative language and accompanying model checker for reasoning about the knowledge and beliefs of multiple agents, and about how those attitudes change when events occur that some agents witness and others do not. *delhi* implements $\mathbf{mB}^+$, a plausibility-model semantics obtained by extending Buckingham’s $\mathbf{mB}$ with the safe-belief and conditional-belief operators of Baltag and Smets, together with conditional effects and a systematic catalogue of derived attitudes. The distinguishing feature of the plausibility setting is that a single preorder per agent generates *both* knowledge and belief: knowledge quantifies over an agent’s whole comparability class, belief over the maxima of its ordering. This yields a system in which an agent may be confidently and coherently wrong—about the world, and, more interestingly, about another agent’s mind—and in which belief revision is a reordering rather than a deletion, so that a lie can be told and later corrected.

We give the syntax and semantics of the formula language, the surface language in which domains are written, and the action layer, in which the modeller declares *who observes what* and the event models are derived rather than hand-built. We formalise a query language that is exactly the formula language extended with a hole, and an enumeration procedure over modal literals that answers questions of the form “what does this agent believe that is false?” rather than requiring the analyst to guess the formula.

Three results are not present in the source literature. First, safe belief *collapses under introspection*: $K_i\Box_i\varphi \equiv K_i\varphi$ and $B_i\Box_i\varphi \equiv B_i\varphi$, so an agent can never certify a safe belief short of knowledge and unconditionally believes its beliefs to be safe—which makes safe belief an analyst’s operator rather than an agent’s. Second, the two product-update rules in the literature diverge on exactly one of four event configurations, and that configuration is realised by the standard demonstration scenario, where one rule prevents the scenario’s own conclusion from arising. Third, bisimulation contraction converts exponential growth into a fixed point on some domains and demonstrably not on others, which constrains any planner built over this semantics; we also measure the incompleteness of the standard contraction by exhaustive enumeration and diagnose its cause. We close by positioning $\mathbf{mB}^+$ against DEL, $\mathbf{mA}^*$, EFP and PDKB, and by stating plainly which components are transcribed from the source literature and which are new.

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1 Introduction

Formal accounts of action have, for the most part, been accounts of fact. A state is an assignment of truth values to propositions; an action transforms one assignment into another; a plan is a sequence of actions leading to an assignment that satisfies a goal. This picture underlies classical planning, most model checking of reactive systems, and the greater part of knowledge representation.

A substantial class of problems does not fit it. In these problems what matters is not only how the world is, but what the agents situated in it take the world to be—and, where several agents are involved, what each takes the others to take it to be. A tutoring system must distinguish a student who is wrong from one who is uncertain, because the remedies differ. A robot working alongside people must know which of its observations its collaborators share. A protocol is secure not when a secret is true but when the adversary does not know it. A negotiator reasons about what the other party believes about their own position. In each case the object of reasoning is an *attitude*—knowledge, belief, uncertainty—held by someone toward a proposition, and the attitude can come apart from the proposition.

Reasoning of this kind is the subject of *epistemic logic*, which supplies modal operators for knowledge and belief and interprets them over structures of alternative possibilities (Fagin et al., 1995), and of *dynamic epistemic logic*, which adds an account of how attitudes change when events occur (Baltag et al., 1998; van Ditmarsch et al., 2007). The theory is mature. What remains difficult is the step from theory to practice: given a scenario, to construct the corresponding structure, compute the effect of an event upon it, and answer questions about the result—mechanically, and at a scale beyond what can be drawn by hand.

This paper concerns that step. Our claim is not that new logical machinery is required, but that a specific and identifiable group of reasoning tasks is poorly served by existing implementations, that the tasks are not exotic, and that supporting them determines a particular set of design choices. We name the tasks first, since they organise everything that follows.

1.1 Five kinds of epistemic reasoning

The following are not five topics but five distinct inferential capacities, in a rough order of increasing demand on the representation. Each is stated in general terms, then illustrated, then paired with the reason it resists naive treatment.

(R1) Attributing an attitude that may be false. To hold that an agent believes φ while φ is false. This is the minimal departure from fact-based reasoning and the one on which the rest depend. It cannot be obtained by adding a proposition `a_believes_p` to a fact-based representation, because nothing then relates that proposition to p : the two vary independently, and every action must maintain the correspondence by hand. Nor can it be obtained from knowledge alone, since knowledge is factive— $K_a\varphi \rightarrow \varphi$ is valid on any standard account, so a false belief is not merely awkward to express but excluded.

(R2) Attributing attitudes about attitudes. To hold that a believes that b believes φ , at arbitrary depth. Deception requires it: to mislead is to bring about a state in which the other party believes something one knows to be false. So do coordination, teaching, and pretence. Developmental psychology treats the capacity as a milestone and tests it with the false-belief tasks of Wimmer and Perner (1983) and Baron-Cohen et al. (1985), whose second-order variants (Sullivan et al., 1994) require an agent to be wrong about *another agent's* mind rather than about the world.

(R3) Deriving differential consequences of a single event. One event, several agents, different epistemic consequences for each. If one agent watches another look inside a container,

the observer learns the contents, the watcher learns that the observer learned them, and a third party who was absent learns nothing—and must continue to regard the pre-event situation as possible. Reasoning of this kind is what makes partial observability epistemic rather than merely probabilistic, and it does not reduce to updating a shared state, because there is no single posterior to update to.

(R4) Revising a belief that is contradicted. To take an agent that believes φ , present it with $\neg\varphi$, and obtain a coherent successor state. Two properties are wanted and are easily lost. Revision should be *local*: correcting an agent about one matter should not resolve its unrelated uncertainties. And it should be *reversible*: an agent misled once must remain able to recover the truth later, which requires that the discarded alternatives still exist in some form. Update rules that eliminate possibilities satisfy neither.

(R5) Assessing the robustness of an attitude. To ask not only whether an agent holds an attitude but whether it would survive further information. A belief that any true statement would overturn and a belief that no true statement would overturn are different objects, and the difference is decisive for anyone deciding whether to intervene: an agent who will discover its error unaided needs no correction, whereas one who will not may need it urgently. Neither knowledge nor belief expresses this, and asking for both returns the same pair of answers in the two cases.

To these five we add a sixth concerning the interface rather than the semantics.

(R6) Searching over attitudes rather than checking them. A question such as *what does this agent believe that is false?* is not a formula to be evaluated but a family of formulas to be searched. Checking answers a question already formulated; diagnosis requires the question to be found. The distinction matters in practice because the interesting attitudes in a model of any size are the ones the analyst did not anticipate.

1.2 A scenario requiring all six

We fix one scenario now and retain it throughout the paper, extending it where a later section needs something it does not yet exercise. It is deliberately small—three agents, two propositions—because the reasoning rather than the size is the difficulty.

An urban search-and-rescue team is working through a partly collapsed building. The members are physically separated, coordinate by radio, and each sees only their own sector. **There is a casualty in the east wing.** Ana swept that wing and found it empty—before the aftershock, so her report was true when she made it and is no longer. Ben is working elsewhere and has no view of that wing. A drone with a thermal camera sees the casualty.

The setting is chosen because the divergences in it are *structural* rather than adversarial. Nobody in the scenario is careless and nobody need be dishonest; beliefs come apart because the team is distributed, radios fail, and information that was accurate becomes stale. That is the ordinary condition of a distributed team, and it is what makes a drone that can reason about its colleagues’ beliefs useful rather than merely clever.

The scenario instantiates each of R1–R6. Ana’s belief that the wing is clear is false, and blamelessly so (R1). If Ben’s radio fails and the drone then reports the casualty, Ana—who did not see the radio fail—will take Ben to have heard it, so she is wrong about *his* state of mind (R2). One radio call reaches Ana, reaches Ben only if his radio is up, and tells the drone nothing it did not already know (R3). Correcting Ana must not incidentally settle her uncertainty about

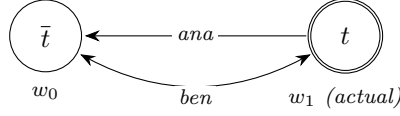


Figure 1: The search-and-rescue team, as `delhi` builds it from the `initially` block of Listing 1. Write t for `trapped`. There is a casualty, so w_1 is actual (double ring). Ana’s arrow points to w_0 : she cannot rule out either world and finds the wrong one more plausible. Ben relates the two worlds in both directions — he has no leaning at all. The drone has no arrow, so it knows. Three agents, one proposition, three different epistemic positions, none of which a fact-only representation can hold.

Ben’s radio, and must leave her open to correction again (R4). Whether Ben would discover the casualty unaided is what determines whether the drone should intervene at all (R5). And the drone’s operative question—*which of my colleagues believes something that is not so?*—is a search rather than a formula (R6).

Two distinctions the scenario draws are worth naming now, because they are easily conflated and the formalism separates them.

Lying and deception are not the same act. To *lie* is to assert what one believes false; Ana does this if she reports the wing clear after being told otherwise. To *deceive* is to bring about a false belief, and that can be done without any lie at all. When Ben’s radio fails and the drone truthfully reports the casualty, Ben is left believing the wing clear—and no one has said anything false. The first is a property of an announcement’s content, the second a property of who received it, and Section 6 shows that `delhi` represents them by different mechanisms.

Belief yields to testimony; knowledge does not. If Ben is told the wing is clear he may come to believe it. If Ben has *been in and looked*, the same assertion leaves him unmoved. This is not a stipulation but a consequence of the semantics, and Section 4 makes it precise.

Listing 1 gives the scenario as a `delhi` domain and Figure 1 the structure derived from it. One line of the domain deserves attention now, because it is where R5 becomes concrete:

```
goal { [] [ben] trapped }
```

The goal is not that Ben believes there is a casualty, nor that he knows it, but that he believes it *stably*: that no further true information would unsettle it. The operator is *safe belief*, treated in Section 4.5. We draw attention to it here for a methodological reason rather than a technical one—a goal of this form cannot be written down at all in a language whose attitudes are knowledge and belief, so the adequacy of a formalism for R5 can be settled by inspection, before any question of efficiency arises.

1.3 Contributions

1. **A formalisation of $\mathbf{mB}^+$** (Section 4), extending $\mathbf{mB}$ (Buckingham, 2021) with safe belief $\Box_i$ and conditional belief B_i^ψ , taken from Baltag and Smets (Baltag and Smets, 2008), together with conditional ontic effects. We are explicit that the operators are not new; what is new is their presence in the object language of an action formalism, where they become queryable properties of a planning state.
2. **A surface language** (Section 5) in which initial states are given *declaratively*—as a list of facts and attitudes that the model must satisfy—rather than as a hand-drawn Kripke structure, together with non-recursive definitions, Horn rules over static constants, and state invariants.

```

types    { Actor - Object }
objects  { ana, ben, drone - Actor }
agents   { ana, ben, drone }
props    { trapped, linked }      // a casualty in the east wing; ben's radio is
                                   working

initially {
    trapped                // there is someone there
    linked                 // ben's radio is up, for now
    B[ana] !trapped        // ana swept the wing pre-aftershock: honest, and
    stale
    ?[ana] trapped        // she has not been back
    ?[ben] trapped        // ben has no view of that wing
}

goal { [][ben] trapped }      // he believes it, and SAFELY

actions {
    radio_fails() {          // ana does not see this happen
        actor ben
        causes !linked
        ben observes, drone observes
    }
    drone_reports() {        // truthful, and carried by radio
        actor drone
        announces trapped
        drone observes
        ana observes
        ben observes if linked
    }
    ana_reports_clear() {    // an honest error before she is told; a lie after
        actor ana
        announces !trapped
        ana observes, drone observes
        ben observes if linked
    }
    ben_enters() {           // sensing: he comes to KNOW
        actor ben
        determines trapped
        ben observes
        drone observes
        ana aware           // she hears him go in, without seeing what he finds
    }
}

```

Listing 1: `search_rescue.delhi`. Sections are explained in Section 5; for now note only that nothing here is a Kripke structure. The modeller states what is true, what is believed, and who observes what — and `delhi` derives Figure 1 and the event models from that.

3. **A query language and enumeration procedure** (Section 7). Queries are formulas of the full modal language extended with a hole $_$; enumeration ranges over modal literals to depth d , following the representation of Muise et al. (2015). Repeated holes are filled uniformly, which is what allows a single pattern to express “what does a believe that is false” as $B_a_ \wedge \neg_$.
4. **An implementation and its measurement** (Sections 8 and 9). We report the effect of bisimulation contraction on iterated update and show that, while it converts exponential growth into a fixed point on several standard domains, there are domains on which it demonstrably does not.
5. **An explicit accounting** (Section 10) of which parts of the system are transcribed from the literature and which are new, including three points at which the source documents disagree with each other and the resolution adopted.

1.4 Scope

delhi is a model checker, not a planner and not a theorem prover. It answers whether a formula holds in a given finite state, and computes the state resulting from a named action. It does not search for a sequence of actions achieving a goal; plan existence in the general dynamic-epistemic setting is undecidable (Bolander and Andersen, 2011), and the restrictions that recover decidability are orthogonal to the semantic questions treated here. Validity across all models is likewise out of scope: every judgement reported is a judgement about a particular pointed model.

Two assumptions run through the whole development and are worth stating before they are used rather than after.

The analyst knows which world is actual. A pointed model designates one world, and the modeller supplies it. Every judgement involving $\Box_i$ depends on that designation, since safe belief is measured from the actual world outward—so $\Box_i$ is available to someone reasoning *about* the scenario and, as Definition 4.11 makes precise, not to an agent reasoning *inside* it. Where the actual world is itself uncertain, the framework as presented does not apply.

Domains are small enough for explicit-state checking. Models are represented extensionally, as sets of worlds with bitset valuations and relations. This is what makes the measurements of Section 9 meaningful and also what bounds them: the approach suits scenarios of the size found in the theory-of-mind and epistemic-planning literature—tens to a few thousand worlds—and not domains requiring symbolic or factored representation.

2 How existing formalisms support these tasks

We now consider the principal families of formalism against R1–R6, using the scenario of Section 1.2 to make each judgement concrete. The aim is not comparative scoring. Each family made a deliberate trade and obtained something for it, and several are more capable than *delhi* in respects this paper does not address—notably plan search. The purpose is to locate where each family ceases to be able to *express* a task, because a formalism that cannot express a task cannot be assessed on how well it performs it, and those expressive boundaries are what determined the design choices set out in Section 2.6.

Table 1 summarises; the subsections give the reasoning.

2.1 Classical planning over facts

A STRIPS-style representation has propositions and actions that change them. It can record `trapped = true`. It cannot record that Ana takes the wing to be clear, because there is no place

	R1	R2	R3	R4	R5	R6
	false belief	higher order	differ- ential	revi- sion	robust- ness	search
Classical planning	—	—	—	—	—	—
Knowledge-only ($S5$; $m\mathcal{A}^*$, EFP)	—	○	✓	—	—	—
Knowledge + belief, two relations	✓	✓	✓	○	—	—
Depth-bounded (PDKB)	✓	○	✓	○	—	○
DEL, event models supplied	✓	✓	✓	✓	○	—
Plausibility DEL (Baltag–Smets)	✓	✓	✓	✓	✓	—
mB⁺ / delhi	✓	✓	✓	✓	✓	✓

Table 1: Expressive coverage of the tasks of Section 1.1. ✓ = expressible; ○ = expressible with a qualification given in the text; — = not expressible. The table concerns expressiveness only, and a dash is not a criticism: restricting to knowledge is part of what buys $m\mathcal{A}^*$ and EFP their planning performance. Note also what the table does *not* have a column for. Plan search would be one, and **delhi** would take a dash in it; so would scale, where the depth-bounded approach leads. The full row of ticks records only that **mB⁺** can state all six tasks, not that it is the best tool for any purpose.

in the representation for a proposition to be true and simultaneously taken to be false. One could add an atom `ana_thinks_clear` and treat it as an ordinary fluent, but then nothing connects it to `trapped`: the two vary independently, no action is constrained by their relationship, and the modeller maintains the correspondence by hand for every action they write. **Stops at:** R1. The mistake is not representable, and R2–R6 all presuppose it.

2.2 Knowledge-only epistemic logic

Give each agent an $S5$ relation and the situation improves markedly. We can now say that Ana does not *know* whether anyone is in the wing, that the drone does know, and that Ben does not — three distinct positions, correctly. Public-announcement dynamics will even update all three when the drone transmits.

What cannot be said is the one thing the scenario is about. Knowledge is *factive*: $K_i\varphi \rightarrow \varphi$ is valid, so $K_{ana} \neg trapped$ is unsatisfiable when the wing is occupied. Ana’s error is not merely hard to encode; it is ruled out by the logic. The best available paraphrase is that she is *ignorant* of whether anyone is there — which is true, and which loses the entire point, since an ignorant searcher would go back and a mistaken one will not.

Stops at: R1, and hence R4 and R5. R3 is well supported—public-announcement dynamics handle differential observability—and R2 is available for knowledge about knowledge, which is why the table records it as qualified rather than absent. This is where $m\mathcal{A}^*$ (Baral et al., 2022) and the EFP planners (Le et al., 2018) sit, and it is a deliberate choice — restricting to knowledge is part of what buys them planning performance.

2.3 Knowledge and belief as two relations

The obvious repair is a second relation per agent: an $S5$ relation for knowledge and a serial, transitive, Euclidean one for belief, with axioms linking them. Now $B_{ana} \neg trapped \wedge trapped$ is satisfiable, and the scenario’s premise is finally expressible.

Two problems remain, and they are R4 and R5.

Revision is destructive. When the drone’s report contradicts Ana’s belief, her belief relation must be rebuilt. The rules available in this setting are blunt: in $m\mathcal{A}^*$ ’s treatment, contradicting information collapses *all* of an agent’s uncertainty, not just the part that conflicted. Ana also does not know whether Ben’s radio is still up; being corrected about the casualty should

not settle that question, and under a collapsing rule it does. Worse, the collapse is irreversible in the wrong way: having been told, she cannot later be brought back by better evidence, because the worlds that would support the alternative are gone.

Stability is inexpressible. The drone needs to know whether Ben *would find out anyway*. With knowledge and belief alone there is no operator for “believes, and no true information would change that”. One can ask whether he believes, and whether he knows, and get *yes* and *no* — and those two answers are the same for a searcher who is about to walk through that wing and one who will never go near it. The goal of Listing 1 is not expressible here.

2.4 Dynamic epistemic logic with hand-built event models

DEL (Baltag et al., 1998; van Ditmarsch et al., 2007) is expressive enough for all of it. Plausibility extensions (Baltag and Smets, 2006, 2008) give belief revision that reorders rather than deletes, and safe belief to state the goal. Nothing in the scenario is beyond it, and *delhi*’s semantics is a DEL semantics.

The cost is where the modeller’s effort goes. In DEL as such, the event model for each action is *supplied*, not derived. To say “the drone transmits, and Ben receives it only if his radio is up” the modeller must construct a multi-event structure with the right preconditions and the right per-agent edges, and get every edge right — and then do it again for the sensing variant, and again for the lie. Section 6.1 shows what those structures look like; they are not hard, but they are fiddly, and an error in one edge produces a model that is well-formed and silently wrong.

There is also no standard *query* apparatus. Having built the model, the drone’s question — *which of my colleagues believes something that is not so?* — is not a formula, it is a search over formulas, and DEL as a logic has nothing to say about how to conduct it.

Stops at: R6, and R5 unless the plausibility extension is adopted. R1–R4 are fully supported, which is why *delhi*’s semantics is a plausibility-DEL semantics rather than an alternative to one. The practical costs are that event models are supplied by the modeller and that plan existence is undecidable in the general setting (Bolander and Andersen, 2011).

2.5 Depth-bounded epistemic states

Bounding nesting depth and compiling to classical planning (Muisse et al., 2015) makes the problem tractable and scales well. Within the bound the representation handles belief, and the modal-literal representation is a genuinely good idea — we borrow it in Section 7.2. But the bound is on the representation, revision is limited, and safe belief is not present, so once again the goal cannot be stated.

Stops at: R5. R2 holds to the depth bound and R4 in a limited form, and the enumeration of R6 is supported in the sense that the modal-literal representation is exactly a search domain—though it is used there to represent states rather than to search over them.

2.6 What the tasks require of a design

Reading Table 1 back along its rows gives a specification, and it is the specification *delhi* was built against. Each item names the task that motivates it and the section that delivers it.

1. **Belief that can be false, with revision that preserves everything else** (R1, R4). Correcting Ana about the casualty must not settle her uncertainty about Ben’s radio, and must leave room for her to be corrected again. This is what plausibility models buy: revision *reorders* worlds instead of deleting them (Section 3).
2. **Safe belief in the object language** (R5). The success criterion is stability, not mere belief, so the language must be able to state it — in a goal, in an invariant, and in a query (Section 4). This is the extension we mark with the +.

3. **Event models derived from declared observability** (R3). The modeller should write `ben observes if linked` and get the right multi-event structure, including the conditional case, rather than draw it (Section 6).
4. **Three observer positions, not two** (R3 at second order, with R2). “Received the call” and “did not receive it” are different, and so is “knew a colleague went in without learning what they found” — which is exactly Ana’s position when Ben enters the wing (Section 6.3).
5. **Search over formulas, not just checking of them** (R6). *Which of my colleagues believes something that is not so?* must be answerable without the analyst guessing the formula first (Section 7).

These five are the paper’s design choices. Each answers a task of Section 1.1 that at least one established family cannot express, and the remaining sections deliver them in that order. Nothing in the list is a new logical idea; the contribution is the combination, and the claim is that the combination is what the six tasks jointly require.

3 Preliminaries

We assume finite sets $\mathcal{P}$ of propositional atoms and $\mathcal{G}$ of agents.

3.1 Plausibility models

The central object is a Kripke structure in which each agent’s accessibility relation is a *preorder* rather than an equivalence, read as a comparative judgement of plausibility.

Definition 3.1 (Plausibility model). A *plausibility model* is a triple $M = \langle W, R, V \rangle$ where

- W is a finite non-empty set of *worlds*;
- $R : W \times W \rightarrow (\mathcal{G} \rightarrow \mathbb{B})$, written $uR_i v$ for $R(u, v)(i) = \text{true}$ and read “agent i considers v at least as plausible as u ”;
- $V : \mathcal{P} \rightarrow 2^W$ is a valuation.

Each R_i is required to be *locally well-preordered*: a preorder (reflexive and transitive) whose restriction to any comparability class is connected. A *state* is a pointed model $\langle M, u \rangle$ with $u \in W$ the *actual* world.

Remark 3.2 (Direction of the ordering). Plausibility *increases* along the arrow: $uR_i v$ says v is the more credible of the two. This convention is worth stating explicitly because a model built with the relation inverted is still a well-formed model—it satisfies every frame condition of Definition 3.1—so no structural validation can detect the error. Every answer the system gives will simply be reversed.

Two relations are derived from R_i , and the whole of the semantics rests on the contrast between them.

Definition 3.3 (Comparability and maxima). For a plausibility model M , agent i , world u and $C \subseteq W$:

$$u \sim_i v \quad \text{iff} \quad uR_i v \text{ or } vR_i u, \quad (\text{comparability}) \quad (1)$$

$$\sim_i^u := \{v \mid u \sim_i v\}, \quad (\text{the class of } u) \quad (2)$$

$$\rightarrow_i C := \{u \in C \mid u'R_i u \text{ for all } u' \in C\}, \quad (\text{maxima of } C) \quad (3)$$

$$\rightarrow_i^u := \rightarrow_i \sim_i^u. \quad (4)$$

Proposition 3.4. $\sim_i$ is an equivalence relation on W .

Proof. Reflexivity and symmetry are immediate from the definition and the reflexivity of R_i . For transitivity, suppose $u \sim_i v$ and $v \sim_i w$. Local connectedness makes R_i total on the comparability class containing u and v , and v and w lie in that same class; hence u and w are comparable. $\square$

Proposition 3.5 (Non-emptiness of maxima). *If C is a non-empty finite subset of a single comparability class, then $\rightarrow_i C \neq \emptyset$.*

Proof. Local connectedness makes R_i restricted to that class a total preorder. A finite non-empty total preorder has maximal elements. $\square$

Remark 3.6 (A precondition that must not be dropped). Proposition 3.5 is stated in Buckingham (2021) without the single-class restriction, and in that generality it is false: if C contains worlds drawn from two distinct comparability classes, no element of either is R_i -related to any element of the other, so $\rightarrow_i C = \emptyset$. Both uses of $\rightarrow_i$ in $\mathbf{mB}^+$ respect the restriction— $\rightarrow_i \sim_i^u$ takes a whole class, and the conditional-belief case of Definition 4.2 intersects a class with a denotation and treats the empty case explicitly—but an implementation that asserts non-emptiness unconditionally will fail on inputs the semantics permits.

3.2 Why one relation and not two

An alternative design gives each agent two relations, one equivalence for knowledge and one serial-transitive-Euclidean relation for belief, and constrains them to interact. The plausibility approach derives both from a single preorder, which has two consequences that matter for this paper.

First, the interaction axioms come for free rather than being imposed. Because $\rightarrow_i^u \subseteq \sim_i^u$ by construction, $K_i\varphi \rightarrow B_i\varphi$ is valid on every model, and no separate frame condition is required to secure it.

Second—and this is the point of the exercise—belief revision becomes a *reordering* rather than a deletion. Learning that ψ does not remove the $\neg\psi$ -worlds from the model; it promotes the ψ -worlds above them. The agent’s knowledge is therefore unchanged by an announcement, while its belief may be overturned, and because the demoted worlds are still present, a subsequent truthful announcement can promote them back. A semantics that deleted worlds would make the first lie irreversible.

4 The logic $\mathbf{mB}^+$

4.1 Syntax

Definition 4.1 (Formula language). The language $\mathcal{L}_{\mathcal{G}}^{\mathcal{P}}$ is generated by

$$\varphi ::= p \mid \neg\varphi \mid \varphi \wedge \varphi \mid K_i\varphi \mid B_i\varphi \mid C_g\varphi \mid B_i^\psi\varphi \mid \Box_i\varphi$$

with $p \in \mathcal{P}$, $i \in \mathcal{G}$, $g \subseteq \mathcal{G}$, and $\psi \in \mathcal{L}_{\mathcal{G}}^{\mathcal{P}}$. We write $\mathcal{L}^{\mathcal{P}}$ for the propositional fragment (the first three clauses). The connectives $\vee$, $\rightarrow$, $\top$ and $\perp$ are defined as usual.

The first six clauses are those of $\mathbf{mB}$ (Buckingham, 2021, Def. 1). The last two, conditional belief and safe belief, are the extension marked by the $+$; they are Baltag and Smets’s operators (Baltag and Smets, 2006, 2008), and were already interpretable in $\mathbf{mB}$ ’s models, since those models *are* plausibility models. What $\mathbf{mB}$ does not do is lift them into its object language. $\mathbf{mB}^+$ does.

4.2 Semantics

Definition 4.2 (Satisfaction). For a state $\langle M, u \rangle$ with $M = \langle W, R, V \rangle$:

$$\langle M, u \rangle \models p \quad \text{iff} \quad u \in V(p) \quad (5)$$

$$\langle M, u \rangle \models \neg\varphi \quad \text{iff} \quad \langle M, u \rangle \not\models \varphi \quad (6)$$

$$\langle M, u \rangle \models \varphi \wedge \chi \quad \text{iff} \quad \langle M, u \rangle \models \varphi \text{ and } \langle M, u \rangle \models \chi \quad (7)$$

$$\langle M, u \rangle \models K_i\varphi \quad \text{iff} \quad \langle M, v \rangle \models \varphi \text{ for all } v \in \sim_i^u \quad (8)$$

$$\langle M, u \rangle \models B_i\varphi \quad \text{iff} \quad \langle M, v \rangle \models \varphi \text{ for all } v \in \rightarrow_i^u \quad (9)$$

$$\langle M, u \rangle \models \Box_i\varphi \quad \text{iff} \quad \langle M, v \rangle \models \varphi \text{ for all } v \text{ with } uR_iv \quad (10)$$

$$\langle M, u \rangle \models B_i^\psi\varphi \quad \text{iff} \quad \langle M, v \rangle \models \varphi \text{ for all } v \in \rightarrow_i(\sim_i^u \cap \llbracket \psi \rrbracket) \quad (11)$$

$$\langle M, u \rangle \models C_g\varphi \quad \text{iff} \quad \langle M, v \rangle \models \varphi \text{ for all } v \text{ with } u \left(\bigcup_{i \in g} \sim_i \right)^* v \quad (12)$$

where $\llbracket \psi \rrbracket = \{v \mid \langle M, v \rangle \models \psi\}$ and $(\cdot)^*$ denotes reflexive-transitive closure. The conditional-belief clause is vacuously true when $\sim_i^u \cap \llbracket \psi \rrbracket = \emptyset$.

Three points deserve comment.

Common knowledge, not common belief. C_g is built on $\sim_i$, the comparability relation, and is therefore common *knowledge*. Some presentations use the same symbol for the closure over R_i , which yields common *belief*; the two are not interchangeable, and **delhi** implements the knowledge reading throughout.

B_i is a special case of B_i^ψ . Taking $\psi = \top$ gives $\sim_i^u \cap W = \sim_i^u$, so $B_i^\top\varphi \equiv B_i\varphi$. The implementation tests this identity rather than assuming it.

Safe belief is anchored to the actual world. K_i quantifies over a fixed set (the class), B_i over another fixed set (the maxima). $\Box_i$ quantifies over $\{v : uR_iv\}$, which depends on *where* u sits in the ordering. This gives $\Box_i$ a character quite unlike the other two, developed in Section 4.5.

4.3 The strength hierarchy

Theorem 4.3 (Hierarchy). *For every model, agent and formula,*

$$K_i\varphi \rightarrow \Box_i\varphi \rightarrow B_i\varphi,$$

and neither implication is reversible.

Proof. The three operators quantify over $\sim_i^u \supseteq \{v : uR_iv\} \supseteq \rightarrow_i^u$. The first inclusion holds because uR_iv implies $u \sim_i v$. The second holds because a maximal element of the class is R_i -above every element of it, u included. A larger quantification domain yields a stronger claim, giving both implications. Failure of the converses is witnessed in Example 4.8. $\square$

Figure 2 is the same statement as a picture, and it is worth keeping in mind for the rest of the paper: the three attitudes differ only in how much of the agent's comparability class they are required to survive.

Theorem 4.4 (Frame properties). *On every plausibility model:*

1. K_i satisfies T , 4 and 5, hence is an $S5$ modality;
2. B_i satisfies D , 4 and 5, hence is a $KD45$ modality, and does not satisfy T ;

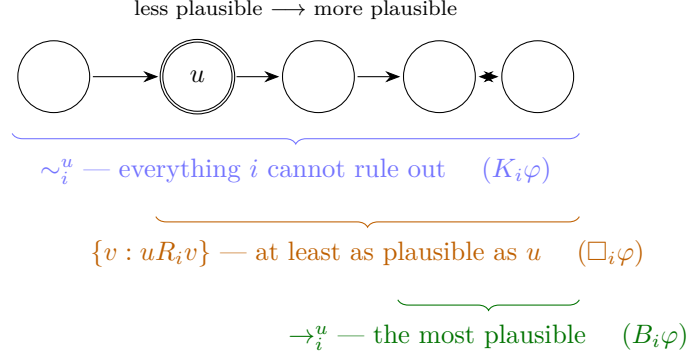


Figure 2: The three attitudes as three nested quantification domains, for one agent i at the actual world u . K_i reads the whole class, B_i only the maxima, and $\Box_i$ the worlds from u upward. Because the middle region is pinned to u , it shrinks as the agent’s belief becomes more accurate—which is the asymmetry developed in Section 4.5.

Surface	Notation	Definition	Reading
$K'[a]\varphi$	$\hat{K}_a\varphi$	$\neg K_a\neg\varphi$	considers possible
$B'[a]\varphi$	$\hat{B}_a\varphi$	$\neg B_a\neg\varphi$	does not rule out
$S'[a]\varphi$	$\hat{S}_a\varphi$	$\neg\Box_a\neg\varphi$	safe-belief dual
$Kw[a]\varphi$	$Kw_a\varphi$	$K_a\varphi \vee K_a\neg\varphi$	knows whether
$Bw[a]\varphi$	$Bw_a\varphi$	$B_a\varphi \vee B_a\neg\varphi$	believes whether
$?[a]\varphi$	$?_a\varphi$	$\neg Kw_a\varphi$	ignorant whether
$??[a]\varphi$	$!_a\varphi$	$\neg Bw_a\varphi$	suspends judgement
$C[*]\varphi$	$C_{\mathcal{G}}\varphi$	$C_g\varphi, g = \mathcal{G}$	common knowledge
$K[a, b]\varphi$	—	$K_a\varphi \wedge K_b\varphi$	agent-list distribution

Table 2: Derived attitudes. Agent lists distribute over K , B , $\Box$, Kw , Bw , $?$ and $??$. Each is a Boolean combination of the primitives, so the evaluator implements eight cases and no more.

3. $\Box_i$ is *factive*: $\Box_i\varphi \rightarrow \varphi$. It satisfies 4 but not 5, and is therefore *S4* rather than *S5*—see Proposition 4.6;
4. $K_i\varphi \rightarrow B_i\varphi$, $B_i\varphi \rightarrow K_iB_i\varphi$, and $K_i\varphi \rightarrow \Box_i\varphi$.

Proof sketch. (1) is immediate from $\sim_i$ being an equivalence relation. (2): seriality follows from Proposition 3.5 applied to $\sim_i^u$, which is non-empty since $u \sim_i u$; introspection follows because $\rightarrow_i^u$ depends only on the class of u , and all worlds in a class have the same class. Failure of T is witnessed by any model in which $u \notin \rightarrow_i^u$. (3): reflexivity of R_i places u in its own quantification domain. (4): the inclusions of Theorem 4.3, plus the class-invariance of $\rightarrow_i^u$ for the second conjunct. $\square$

Remark 4.5. The asymmetry in (2)—consistency but not factivity—is the formal content of the informal claim that an agent may be *wrong but not incoherent*. $B_i\varphi \wedge \neg\varphi$ is satisfiable; $B_i\varphi \wedge B_i\neg\varphi$ is not.

4.4 Derived attitudes

The implemented core has exactly the eight cases of Definition 4.2. Every further attitude in Table 2 is defined by translation and eliminated before evaluation.

The “whether” operators are worth singling out. $Kw_a\varphi$ is not $K_a\varphi$ and not its negation: it is a *disjunction* of two knowledge claims, asserting that the agent has settled the question without

saying which way. Questions about ignorance are its negation. That these require a disjunction is what makes them operators rather than abbreviations a user can be expected to expand by hand, and Section 7 shows that the same disjunctive structure is what lets the enumeration procedure reach them.

4.5 Safe belief

Safe belief occupies the gap in Theorem 4.3, and because it is measured from the actual world its behaviour is asymmetric in a way worth developing.

4.5.1 Background: safe belief in Baltag and Smets

The operator is not ours and neither is the name. Baltag and Smets (2006, 2008) introduce *safe belief* on plausibility models as part of a programme separating knowledge, belief and plausibility and giving each its own dynamics. Three features of their treatment matter for how we use it, and the third is the one most easily missed.

It is offered as a notion of knowledge, not as a strong belief. Baltag and Smets present safe belief as a *weak, non-negatively-introspective* kind of knowledge—an alternative to the *S5* reading rather than a way-station between *S5* and *KD45*. The motivation is the defeasibility tradition of Lehrer and Paxson (1969) and Stalnaker (2006): knowledge as true belief that no further truth would overturn, which is exactly Proposition 4.9.

Its logic is axiomatised alongside conditional belief. Their work gives complete axiomatisations for knowledge, conditional belief and safe belief over plausibility models, and develops the dynamics—how each attitude moves under action-priority update—together with quantitative refinements such as degrees of safety.

It is *S4*, not *S5*. This is the technical content of “non-negatively introspective”, and it separates safe belief sharply from both of its neighbours.

Proposition 4.6 (Safe belief is *S4*). $\Box_i$ is a normal modality satisfying *T* and 4 but not 5: $\Box_i\varphi \rightarrow \varphi$ and $\Box_i\varphi \rightarrow \Box_i\Box_i\varphi$ are valid, while $\neg\Box_i\varphi \rightarrow \Box_i\neg\Box_i\varphi$ is not.

Proof. Normality is immediate, $\Box_i$ being a box over R_i ; *T* follows from reflexivity and 4 from transitivity (both appear in Theorem 4.11). For the failure of 5, take $W = \{w_0, w_1\}$ with $\llbracket p \rrbracket = \{w_1\}$, $w_0 R_i w_1$, and w_0 actual—an agent whose most plausible world satisfies p while the actual world does not. At w_0 the up-set is $\{w_0, w_1\}$ and p fails at w_0 , so $\neg\Box_i p$ holds. But w_1 is maximal, so its up-set is $\{w_1\}$, where p holds; hence $\Box_i p$ holds at w_1 , $\neg\Box_i p$ fails there, and $\Box_i\neg\Box_i p$ fails at w_0 . $\square$

Remark 4.7 (Where the three attitudes part company). In the model of Proposition 4.6, K_i and B_i both satisfy 5 and $\Box_i$ does not, so the failure is a property of safe belief and not of the model. This is the precise sense in which safe belief is a *weaker* knowledge: it keeps factivity, which belief lacks, and gives up negative introspection, which knowledge has. An agent with $\Box_i$ need not be aware of the boundaries of its own safe beliefs—and Section 4.5.2 establishes the stronger fact that it cannot determine them at all.

What $\mathbf{mB}^+$ adds. The operator, its characterisation and its axiomatisation are Baltag and Smets’s. What we contribute is its *availability inside an action formalism*. In $\mathbf{mB}$ and its relatives the plausibility models already interpret $\Box_i$ and B_i^ψ , but the object language of the action description does not mention them, so they cannot appear in a goal, an invariant, or a query. Lifting them makes safe belief a property one can *check of a planning state*, which in turn makes two questions askable that were not before: which actions confer it (Section 4.5.3), and whether an agent can establish it of itself (Section 4.5.2). The answer to the second is the negative result of Theorem 4.11, and it is not in the source literature.

Example 4.8 (Three positions on one question). Consider $\mathcal{P} = \{up\}$ and three agents, none of whom knows the answer:

```
initially {
  up                                // the server really is up
  ?[ada] up   B[ada] up             // ada cannot tell, but leans the right way
  ?[ben] up   B[ben] !up            // ben cannot tell, and leans the wrong way
  ?[cleo] up                                // cleo has no leaning either way
}
```

Then $B_{ada} up$ and $B_{ben} \neg up$ both hold, so belief cannot separate the first two agents. Safe belief separates all three: $\Box_{ada} up$ holds, $\Box_{ben} \neg up$ does not (indeed cannot, by factivity), and $\Box_{cleo} up$ does not, because both worlds are maximal for *cleo* and so both lie above the actual one.

Proposition 4.9 (Defeasibility characterisation). $\Box_i \varphi$ holds at $\langle M, u \rangle$ if and only if $B_i^\psi \varphi$ holds at $\langle M, u \rangle$ for every ψ with $\langle M, u \rangle \models \psi$.

Proof. ($\Rightarrow$) Assume $\Box_i \varphi$ at u , so φ holds throughout $U := \{v : uR_i v\}$. Let ψ be true at u and let $v \in \rightarrow_i (\sim_i^u \cap \llbracket \psi \rrbracket)$. Since $u \in \sim_i^u \cap \llbracket \psi \rrbracket$, that set is non-empty, and by maximality of v within it we have $uR_i v$. Hence $v \in U$ and φ holds at v . As v was arbitrary, $B_i^\psi \varphi$ holds.

($\Leftarrow$) Contrapositive. Suppose $\Box_i \varphi$ fails, so there is w with $uR_i w$ and φ false at w . Take ψ to be true exactly at w and at the worlds of $\sim_i^u$ ranked no higher than w — concretely, $\llbracket \psi \rrbracket \cap \sim_i^u = \{v \in \sim_i^u : vR_i w\}$, which is expressible whenever the model is finite and worlds are distinguished by atoms, and which contains u because $uR_i w$. Then ψ is true at u , and w is R_i -maximal in $\sim_i^u \cap \llbracket \psi \rrbracket$ by construction, so $w \in \rightarrow_i (\sim_i^u \cap \llbracket \psi \rrbracket)$. Since φ fails at w , $B_i^\psi \varphi$ fails. $\square$

Remark 4.10 (Reading the characterisation). This is the property that makes $\Box_i$ more than a technical interpolant: a safe belief is one that no true information can overturn, since conditioning on a true ψ is exactly what a truthful announcement of ψ does to the ordering. The restriction to true ψ is not decoration — a safe belief may still be overturned by a falsehood, and Section 4.5.3 exhibits a case. The ($\Leftarrow$) direction requires that the separating ψ be expressible, which holds in the finite, atom-distinguished models **delhi** constructs; over models where worlds are not separated by any formula the implication may fail, and $\Box_i$ is then strictly stronger than the conditional-belief conjunction.

4.5.2 Introspection

Because $\Box_i$ is anchored to the actual world, and the actual world is precisely what an agent cannot locate, safe belief is not introspectible in the way knowledge and belief are.

Theorem 4.11 (Introspection collapse). *On every plausibility model:*

$$K_i \Box_i \varphi \equiv K_i \varphi, \quad B_i \Box_i \varphi \equiv B_i \varphi, \quad \Box_i \Box_i \varphi \equiv \Box_i \varphi.$$

Proof. First. ($\Leftarrow$) If φ holds throughout $\sim_i^u$ then in particular it holds throughout every up-set contained in that class, so $\Box_i \varphi$ holds at every $v \in \sim_i^u$. ($\Rightarrow$) Suppose $K_i \Box_i \varphi$ at u , and let v_0 be a minimal element of $\sim_i^u$, which exists because the class is finite and, by local connectedness, totally preordered. Then $\{w : v_0 R_i w\} = \sim_i^u$, and $\Box_i \varphi$ at v_0 gives φ throughout the class, i.e. $K_i \varphi$ at u .

Second. Let $v \in \rightarrow_i^u$. Since v is maximal in its class, $\{w : v R_i w\} = \rightarrow_i^u$. Hence $\Box_i \varphi$ at v iff φ holds throughout $\rightarrow_i^u$, which is $B_i \varphi$ at u ; and this is independent of the choice of v .

Third. Transitivity of R_i gives $\Box_i \varphi \rightarrow \Box_i \Box_i \varphi$; reflexivity gives the converse. $\square$

Corollary 4.12 (Overconfidence). *Every agent believes each of its beliefs to be safe: $B_i \varphi \rightarrow B_i \Box_i \varphi$ is valid, including when $\Box_i \varphi$ is false.*

Route	Instance	K_i	$\Box_i$	B_i
Sensing	the agent looks	✓	✓	✓
Truthful announcement	the agent is told, and it is true	—	✓	✓
Lie	the agent is told a falsehood	—	—	✓

Table 3: Acquisition of the three attitudes toward the announced proposition. The middle row is the case that motivates the operator: a truthful announcement upgrades a previously mistaken belief to a safe one without conferring knowledge.

Proof. Immediate from the second equivalence of Theorem 4.11, which gives $B_i\Box_i\varphi \equiv B_i\varphi$ and hence the implication in both directions. That the consequent may hold while $\Box_i\varphi$ is false is witnessed in Example 4.8: $B_{ben}\Box_{ben}\neg up$ holds because $B_{ben}\neg up$ does, while $\Box_{ben}\neg up$ fails by factivity, $\neg up$ being false at the actual world. $\square$

Corollary 4.12 is not a defect of the semantics but a consequence of belief being evaluated at the maxima: from the top of one’s own ordering, every belief looks unshakeable. In Example 4.8, $B_{ben}\Box_{ben}\neg up$ holds while $\Box_{ben}\neg up$ does not; *ben* takes his belief to be immune to revision at the moment before one true announcement overturns it.

The practical consequence for modelling is that $\Box_i$ is an *external* operator. It measures the fit between an agent’s ordering and the actual world; the agent has no access to the second argument, and its self-report is uninformative because it is unconditionally affirmative. An analyst may ask $\Box_i\varphi$ of a state; an agent may not ask it of itself.

Remark 4.13 (Why the operator is worth having at all). K_i here is *S5*: infallible certainty, true in every world the agent cannot rule out. A long tradition in epistemology holds that this is too strong for the ordinary word *knows*, and proposes instead that knowledge is true belief that no further truth would overturn—the *defeasibility analysis* of Lehrer and Paxson (1969), developed by Stalnaker (2006). By Proposition 4.9 that analysis is exactly $\Box_i$. Safe belief is therefore not a technical interpolant wedged between two real operators; it is a serious candidate for what knowing *is*, and $\mathbf{mB}^+$ carries both readings in one model so that a modeller may ask for either and compare them.

4.5.3 How safe belief is acquired

An agent safely believes φ exactly when the actual world is ranked at least as high as every world where φ fails. Table 3 records the three dynamic routes, anticipating the action layer of Section 6.

Two observations. First, the middle row describes a state with no compact name in ordinary usage: the agent is now right, no true information can dislodge it, and it still cannot rule out the alternative. Second, the bottom row cannot come out otherwise, since $\Box_i$ is factive—but a lie can nevertheless *destroy* a safe belief, by reordering the worlds of an agent that previously held one. Safety is immunity to truth, not permanence.

5 The surface language

A *delhi* domain is a text file with a fixed section structure. We give the grammar, then treat the sections whose design is not immediate.

5.1 Structure

```

types      { Actor - Object }           // type hierarchy
objects    { alice, bob - Actor }       // typed constants
agents     { alice, bob }               // which objects have minds
props      { h, d }                    // propositional atoms
constants  { adjacent(hall, study) }    // static, world-independent
define     { ... }                     // named formulas, expanded away
rules      { ... }                     // Horn clauses over constants
initially  { ... }                     // declarative initial state
state      { ... }                     // ...or an explicit model
goal       { phi }                     // optional
invariants { phi ... }                 // must hold in every state
actions    { ... }                     // action theory

```

Listing 2: Overall file structure. `initially` or `state` is required, as is `actions`, which may be empty.

5.2 Declarative initial states

The design decision most visible to a user is that initial states are given as *constraints* rather than as structures. An `initially` block lists facts and attitudes; `delhi` constructs a model satisfying them and then verifies the model against every line written.

```

initially {
  h                // a fact about the actual world
  ?[carol] h       // carol cannot tell whether h
  B[carol] h       // but she correctly learns that way
}

```

Anything not mentioned is known by every agent, so the block above yields a two-world model in which *carol* is uncertain and the others are not. The alternative form, `state`, writes the model out directly and is what the tool prints when asked to display one:

```

state {
  *w1 <- { h }      // '*' marks the designated world
  w0 <- { }
  carol: w0 < w1    // w1 is the more plausible
}

```

Here `<` and `<=` point toward the more plausible world and `~` relates two worlds in both directions. Agents with no line distinguish every world, which is to say they know everything—the degenerate case of Definition 3.1 in which R_i is the identity.

5.3 Definitions, rules and invariants

Three optional sections address needs that arise in practice and that the underlying semantics does not itself provide.

Definitions. Named formulas, parameterised over objects, expanded before lowering so that the semantics never encounters the name:

```

define {
  blocked(?r)      = !lit(?r) | locked(?r)
  can_enter(?w, ?r) = !blocked(?r) & K[?w] !blocked(?r)
}

```

```
}

```

Parameters may occupy agent positions, as `?w` does above. Cycles are rejected when the definition table is built rather than caught by a depth limit at expansion time. Two restrictions are deliberate: a defined name may not appear as the target of `causes` or as a world fact, since both require an atom the semantics can *set*; and parameters range over objects, not formulas, so second-order macros such as `define f(?p) = K[a] ?p` are not admitted.

Rules. Horn clauses over the static constant table, saturated to a least fixpoint at parse time:

```
constants { !adjacent(Room, Room)  adjacent(hall, study)  adjacent(study, attic) }
rules {
  reach(?x, ?y) :- adjacent(?x, ?y)
  reach(?x, ?z) :- adjacent(?x, ?y), reach(?y, ?z)
}
```

Derived predicates never become propositions, and therefore consume no bit in any world; `reach(hall, attic)` folds to $\top$ exactly as a declared constant would, and ground actions whose guards fold to $\perp$ are discarded before construction.

The restriction to *constants* is the substantive one. The fixpoint is computed once, which is sound only because the extension is world-independent. A rule over a fluent would have an extension varying per world and per action, so evaluating it would require either a fixpoint per world at query time or the maintenance of derived atoms through product update—the frame problem in a new place. `delhi` rejects such rules with a diagnostic rather than supporting them partially. Bodies carry no negation, keeping the program monotone so that the least fixpoint exists, and every head variable must occur in the body, so that no head asserts more than the body justifies.

Invariants. Constraints required to hold in *every* state rather than only the first:

```
invariants { !((B[a] p & B[b] !p) | (B[a] !p & B[b] p)) }
```

An `initially` entry that drives no construction is already an assertion about the start; an invariant is the same assertion made about the whole run, which is usually what a domain constraint is intended to mean. Invariants are checked when the initial state is built and after every action.

6 Dynamics

An action changes a state by product update with an event model. The distinguishing feature of the action-language approach, inherited from `mB` and $m\mathcal{A}^*$, is that the modeller does not construct event models: they declare *what changes* and *who observes*, and the event model is derived. This section explains what is being derived before showing how.

6.1 Background: event models and product update

The machinery is due to Baltag et al. (1998) and is standard in dynamic epistemic logic, but it is not obvious on first encounter, so we develop it from the idea rather than from the tuple.

The idea. An agent can be uncertain about *how the world is*—that is what Figure 1 draws. It can equally be uncertain about *what just happened*. The radio crackles; was that the drone reporting a casualty, or reporting none? Ben moves toward the east wing; did he go in and look,

or turn back? These are uncertainties of the same shape, and the insight behind event models is to represent them with exactly the same apparatus.

So an event is modelled the way a world is: as one possibility among several, with a relation per agent saying which of them that agent cannot tell apart. The correspondence is complete, term for term:

State model $\langle W, R, V \rangle$	Event model $\langle E, Q, pre, add, del, \Gamma \rangle$
<i>worlds</i> W — ways things could be	<i>events</i> E — things that could have happened
<i>valuation</i> V — what is true in a world	<i>precondition</i> pre — when an event can occur; <i>add</i> , <i>del</i> — what it changes
R_i — worlds i cannot separate, ordered by plausibility	$Q(\cdot, \cdot)(i)$ — events i cannot separate, ordered the same way
one <i>designated</i> world — how things actually are	<i>designated</i> events Γ — what actually happens

Table 4: An event model is a state model with the parts renamed. This is the whole conceptual content of the construction; everything after it is bookkeeping.

Why preconditions and not just effects. An event needs to say *when* it can occur, for two reasons. First, actions have genuine requirements—one cannot open a box that is already open. Second, and less obviously, an event’s precondition is what carries *information*: the event “Carol announces φ ” has precondition φ , so learning that this event occurred is precisely learning φ . Sensing and announcing change no atom at all; their entire effect is achieved through preconditions and the edges between events.

Why there is an event in which nothing happens. Every construction in Table 6 includes an event $e^\top$ with precondition $\top$, no effects, and no designation. It exists for the *oblivious* agent. If Alice does not notice that Carol peeked, then after the peek Alice must still consider possible a world in which nothing occurred—and there must be an event for that world to be paired with. Remove $e^\top$ and obliviousness becomes inexpressible: every agent would at least learn that *something* took place.

Product update. The two models are then combined. A world of the new state is a pair $\langle u, e \rangle$, read “the world was u and event e occurred”. Two constraints shape the result:

- **Only consistent pairs survive.** $\langle u, e \rangle$ exists only if $pre(e)$ holds at u . This is where announcements do their work—an announcement of φ has a φ -event and a $\neg\varphi$ -event, and each survives only at worlds where its precondition holds, so the pairing sorts worlds by what was said.
- **Uncertainty multiplies.** Agent i confuses $\langle u, e \rangle$ with $\langle v, f \rangle$ only if it confused u with v and e with f . An agent that can tell the events apart therefore ends up with less uncertainty than it began with, which is what learning amounts to here.

The size consequence is immediate and matters for Section 9: the product has up to $|W| \times |E|$ worlds, so iterating actions multiplies model size, and something must be done about it. That something is Section 8.2.

One refinement, which is where plausibility enters. The account above is the classical one, in which relations are equivalences and the second constraint is the whole story. On plausibility models the events are also *ordered*, and the order of the events takes precedence over the order of the worlds: if the event model ranks e strictly above f , that ranking survives into the product regardless of how the agent previously ranked the underlying worlds. This is *action priority* (Baltag and Smets, 2008), and it is what makes an announcement capable of overturning a prior belief. Definition 6.5 states it precisely and Figure 4 shows it operating; the reader who has followed this far can read both directly.

6.2 Action theories

Definition 6.1 (Action theory). An action theory T for action a is a set of statements of the following forms, with the indicated formula typing:

#	Form	Typing	Role
1	a requires φ	$\varphi \in \mathcal{L}^{\mathcal{P}}$	precondition
2	a causes $l_0, \dots, l_n$ if φ	l_j literals, $\varphi \in \mathcal{L}^{\mathcal{P}}$	ontic effect
3	a determines φ	$\varphi \in \mathcal{L}^{\mathcal{P}}$	sensing
4	a announces ψ	$\psi \in \mathcal{L}_G^{\mathcal{P}}$	announcement
5	i observes a if φ	$\varphi \in \mathcal{L}^{\mathcal{P}}$	full observer
6	i aware_of a if φ	$\varphi \in \mathcal{L}^{\mathcal{P}}$	partial observer

An action theory is *well-formed* when it contains exactly one statement of form 2, 3 or 4; at most one of form 1; no **causes** list containing both p and $\neg p$; and, for every state, no agent satisfying both an **observes** and an **aware_of** condition.

Remark 6.2 (Typing is load-bearing). Form 4 admits a *modal* ψ , which need not be true. This is what makes announcements capable of being lies, and of being about beliefs rather than facts. It also propagates: announcement event preconditions are $a_{pre} \wedge \psi$ and are therefore modal, so precondition evaluation requires full modal entailment against the pre-update model. Edge conditions, by contrast, remain propositional, being Boolean combinations of observability conditions—a distinction worth preserving in an implementation, since edge conditions are evaluated at two worlds per edge per agent and dominate the cost of update.

Remark 6.3 (Preconditions: a divergence in the sources). The source documents give three incompatible readings of multiple form-1 statements: disjunction (Buckingham, 2021, eq. 5.1), at most one statement, and—in the reference implementation’s corpus—conjunction. Under the disjunctive reading, *adding* a precondition makes an action *more* applicable, contradicting both the ordinary meaning of the word and every domain file in that corpus. *delhi* adopts the at-most-one rule: a single **pre** clause per action, conjunction written explicitly, and $a_{pre} = \top$ in its absence. A second **pre** clause is a diagnostic rather than a silent combination.

6.3 Observability

Each agent occupies exactly one of three positions with respect to an action. The three are distinguished by two edge labels, defined from the observability conditions of forms 5 and 6: Write $O_i := \{\varphi \mid (i \text{ **observes** } a \text{ **if** } \varphi) \in T\}$ for the conditions under which i fully observes, and $A_i := \{\varphi \mid (i \text{ **aware_of** } a \text{ **if** } \varphi) \in T\}$ for those under which it is merely aware. Then

$$FPN(i) := \top \quad (13)$$

$$PN(i) := \neg \bigvee O_i \quad (14)$$

$$N(i) := \neg \bigvee (O_i \cup A_i) \quad (15)$$

Position	Declaration	Learns	Labels
Full observer	i observes a	what happened, outcome included	$PN(i) = \perp, N(i) = \perp$
Partial observer	i aware a	that it happened, not how	$PN(i) = \top, N(i) = \perp$
Oblivious	neither	nothing	$PN(i) = \top, N(i) = \top$

Table 5: The three observer positions, with the edge labels each induces (for unconditional declarations). PN governs whether the agent can distinguish the outcome events; N governs whether it can distinguish them from the null event.

Kind	Construction
Ontic	$E = \{e^c, e^\top\}$; $Q(e^c, e^\top) = N$; $pre(e^c) = a_{pre}$, $pre(e^\top) = \top$; $add(e^c) = P^+$, $del(e^c) = P^-$; $\Gamma = \{e^c\}$. With conditional effects, e^c splits into one event per realisable outcome with mutually exclusive preconditions.
Announcement	$E = \{e^\varphi, e^{\neg\varphi}, e^\top\}$; $Q(e^\varphi, e^{\neg\varphi}) = PN$, $Q(e^{\neg\varphi}, e^\varphi) = FPN$, $Q(e^\varphi, e^\top) = Q(e^{\neg\varphi}, e^\top) = N$; $pre(e^\varphi) = a_{pre} \wedge \varphi$, $pre(e^{\neg\varphi}) = a_{pre} \wedge \neg\varphi$, $pre(e^\top) = \top$; $add = del = \emptyset$; $\Gamma = \{e^\varphi, e^{\neg\varphi}\}$.
Sensing	As announcement, but $Q(e^{\neg\varphi}, e^\varphi) = PN$ rather than FPN .

Table 6: The three event-model constructions. P^+ and P^- are the positive and negative literals of the **causes** list.

with the empty disjunction being $\perp$, so an agent named in no clause has $PN(i) = N(i) = \top$.

Table 5 makes precise what is otherwise easy to state loosely. A partial observer retains the edge between the two outcome events—so the outcomes remain indistinguishable to it—and loses the edge to the null event, so it does rule out that nothing happened. That combination is exactly the informal reading “knows the action occurred, not how it turned out”. The consequence is second-order: if j senses φ and i is a partial observer, i comes to know that j knows *whether* φ , without learning φ .

Both declarations admit a condition, evaluated in the state at the time of the action. This is what Buckingham (2021) calls *local dynamic observability*, and it is the mechanism by which an agent’s observer position may change during a trace—for instance, an agent that is a partial observer only while not distracted.

6.4 Event models and the three constructions

Definition 6.4 (Action plausibility model). An *action plausibility model* is $\langle E, Q, pre, add, del, \Gamma \rangle$ with E a finite set of events, $Q : E \times E \rightarrow (\mathcal{G} \rightarrow \mathcal{L}^P)$ the edge conditions, $pre : E \rightarrow \mathcal{L}_G^P$, $add, del : E \rightarrow 2^P$, and $\Gamma \subseteq E$ designated. Every event carries a reflexive FPN edge for every agent; unlisted edges are $\perp$.

The single-cell difference between announcement and sensing in Table 6 carries the whole distinction between being told and finding out, and Figure 3 draws it. With $FPN = \top$ on the backward edge, even a full observer of an announcement retains an edge from $e^{\neg\varphi}$ to e^φ , and so cannot rule out that what was said was false; the result is belief. With PN there, a full observer distinguishes the two events and comes to *know* whether φ .



Figure 3: The announcement and sensing event models differ in one edge label, shown in red. For a full observer $PN = \perp$ and $N = \perp$, so under sensing that observer keeps no edge at all between e^φ and $e^{\neg\varphi}$ and comes to know which occurred; under announcement the $FPN = \top$ edge survives, leaving the possibility that the speaker lied. Reflexive FPN edges are present at every event and omitted from the drawing.

6.5 Product update

Definition 6.5 (State transition). Write $e \rightarrow^{iuv} f$ for $\langle M, u \rangle \models Q(e, f)(i)$ and $\langle M, v \rangle \models Q(e, f)(i)$. The updated model $M' = \langle W', R', V' \rangle$ is

$$W' = \{ \langle u, e \rangle \mid u \in W, e \in E, \langle M, u \rangle \models \text{pre}(e) \} \quad (16)$$

$$R'(\langle u, e \rangle, \langle v, f \rangle)(i) = \begin{cases} (e \rightarrow^{iuv} f) \wedge \neg(f \rightarrow^{iuv} e) \wedge u \sim_i v, & \text{or} \\ (e \rightarrow^{iuv} f) \wedge (f \rightarrow^{iuv} e) \wedge u R_i v \end{cases} \quad (17)$$

$$V'(\langle u, e \rangle) = (V(u) \cup \text{add}(e)) \setminus \text{del}(e) \quad (18)$$

with designated world $\langle d, e \rangle$ for the unique $e \in \Gamma$ whose precondition holds at d .

This is Baltag–Smets action-priority update: event plausibility overrides state plausibility, so incoming information takes precedence over prior belief except where it contradicts prior knowledge.

6.5.1 The update worked through

Figure 4 applies Definition 6.5 to the running example: the initial model of Figure 1 crossed with the announcement model of Figure 3 for $\psi = t$, which is the drone reporting the casualty.

The reversal in Figure 4 is action priority doing its work. The event order says $e^{\neg h}$ is at least as plausible as e^h and not conversely, so the first disjunct of Definition 6.5 fires and the event order *overrides* the prior world order.

Proposition 6.6 (The two update rules diverge, and the divergence is reachable). *Let R_{alt} be the rule*

$$R_{\text{alt}}(\langle u, e \rangle, \langle v, f \rangle)(i) = (u R_i v \wedge (e \rightarrow^{iuv} f \vee f \rightarrow^{iuv} e)) \vee (v R_i u \wedge e \rightarrow^{iuv} f \wedge \neg(f \rightarrow^{iuv} e)).$$

Then R_{alt} and Definition 6.5 agree on three of the four event-comparability configurations and differ exactly when $f \rightarrow^{iuv} e$ holds and $e \rightarrow^{iuv} f$ does not. That configuration is realised by any announcement to a full observer holding a contrary prior belief—in particular by `drone_reports()` in Listing 1—and the two rules then give contradictory answers to whether the report lands.

Proof. Case analysis on the two edge predicates:

Configuration	Definition 6.5	R_{alt}	Agree?
$e \rightarrow f$ and $f \rightarrow e$	$u R_i v$	$u R_i v$	yes
$e \rightarrow f$, not $f \rightarrow e$	$u \sim_i v$	$u R_i v \vee v R_i u = u \sim_i v$	yes
neither	$\perp$	$\perp$	yes
$f \rightarrow e$, not $e \rightarrow f$	$\perp$	$u R_i v$	no

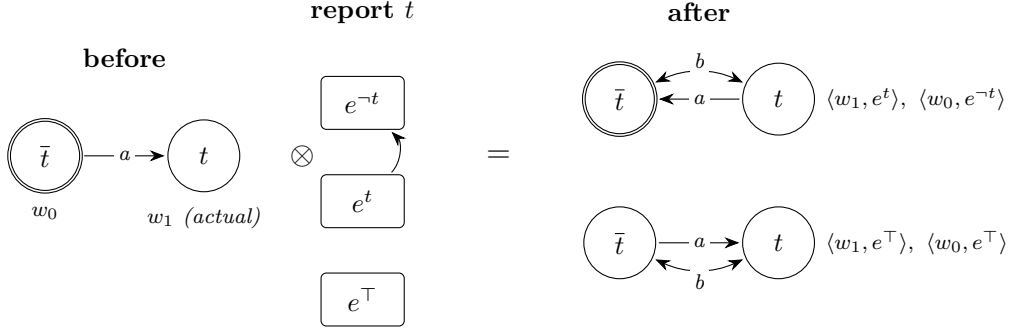


Figure 4: The drone reporting the casualty, worked through; t abbreviates **trapped** and a, b are Ana and Ben. Each original world pairs with the designated event whose precondition it satisfies, and with $e^\top$, giving four worlds in two clusters. **Upper cluster (the call went out)**. The event order makes e^t strictly the more plausible, and by action priority that overrides Ana’s prior order: her arrow has *reversed*, so B_{at} now holds. **Lower cluster ($e^\top$, nothing observable happened)**. Her original order survives untouched — this is the copy that exists for an agent who might have missed the transmission, and it is why $e^\top$ is in the construction at all. Ana is a full observer here, so she has no edge between the clusters; Ben, whose radio is up in this run, likewise receives it. Had his radio failed he would have been oblivious, and the lower cluster would have been where he in fact was — which is the divergence of Section 1.2. Ana’s *knowledge* is untouched throughout, which is what leaves room for a later sighting to upgrade her to knowledge.

For reachability, take $i = ana$, $u = w_0$, $v = w_1$, $e = e^t$, $f = e^{-t}$ in Figure 4. Ana is a full observer, so $PN(ana) = \neg\top = \perp$ and $Q(e^t, e^{-t})(ana) = \perp$, giving $\neg(e \rightarrow f)$; while $Q(e^{-t}, e^t)(ana) = FPN = \top$, giving $f \rightarrow e$. Also $w_1 \sim_i w_0$ holds, $w_1 R_i w_0$ fails, and $w_0 R_i w_1$ holds. Under Definition 6.5 no edge is created, Ana retains a strict preference for the reported world, and B_{anat} holds. Under R_{alt} the first disjunct fires, an edge appears in both directions, the two worlds become equiplausible, and B_{anat} fails. $\square$

Remark 6.7 (Why this settles the choice). The divergence is not a corner case requiring construction. Under R_{alt} the drone’s report does not land: Ana ends up with no belief either way rather than with the true one, so the goal of Listing 1 is unreachable and task R5 cannot be discharged at all.

The same configuration arises in the *Coin Lie* scenario of Buckingham (2021), which the source literature uses precisely to *demonstrate* second-order false belief—so under R_{alt} that demonstration fails on its own example. That is the historical reason to regard Definition 6.5 as authoritative, and it is independent of our scenario. `delhi` implements Definition 6.5; both rules are implemented behind a feature flag and the configuration above is pinned by a regression test.

7 The query language

7.1 Checking

A query is a formula of $\mathcal{L}_G^P$ extended with a hole. Checking is satisfaction:

$$\text{eval}(s, \varphi) = s \models \varphi, \quad \varphi \in \mathcal{L}_G^P,$$

total and decidable, since $\models$ is evaluated over a finite model with memoisation.

```
query    ::= formula
formula ::= atom | 'true' | 'false' | '_'
```

```

| '!' formula
| formula ('&' | '|' | '->') formula
| modality '[' agents ']' formula
| 'B' '~' formula '[' agents ']' formula
| '(' formula ')'
modality ::= 'K' | 'B' | '[]' | 'C' | "K'" | "B'" | "S'"
| 'Kw' | 'Bw' | '?' | '??'
agents   ::= '*' | name (',' name)*

```

Listing 3: Query grammar. Precedence, loosest to tightest: $\rightarrow$ (right-associative), $|$, $\&$, $!$, modality.

The query language is *exactly* the formula language plus $_$: every operator of Definition 4.2 and every derived attitude of Table 2 is available, closed under the Boolean connectives at arbitrary depth. Goals, invariants and interactive queries share one parser and one lowering path, so nothing expressible as a goal is inexpressible as a query.

Remark 7.1 (The hole is a token). $_$ lexes as a distinct token; $_x$ and at_park remain ordinary identifiers. This is load-bearing rather than cosmetic. Filling by textual substitution tears any identifier containing an underscore, turning $_ \& \text{at_park}$ into a malformed formula. Filling is a tree substitution, which additionally means a filled pattern requires no defensive parentheses: $\neg_$ negates whatever fills it regardless of the filler’s precedence.

7.2 Enumeration

Checking answers a question the analyst already knows to ask. The complementary operation enumerates the formulas of a given *shape* that hold.

Definition 7.2 (Ask). For a state s , a pattern π containing at least one hole, and a depth d ,

$$\text{ask}(s, \pi, d) = \{ \pi[\varphi/_] \mid \varphi \in ML_d, s \models \pi[\varphi/_] \},$$

where every hole in π receives the *same* φ and ML_d is the set of *modal literals* of depth at most d :

$$Lit = \{ p, \neg p \mid p \in \mathcal{P} \} \tag{19}$$

$$ML_0 = Lit \tag{20}$$

$$ML_k = ML_{k-1} \cup \{ O_i \varphi \mid O \in \{K, B\}, i \in \mathcal{G}, \varphi \in ML_{k-1} \setminus ML_{k-2} \}. \tag{21}$$

Hence $|ML_d| = \sum_{k \leq d} (2|\mathcal{G}|)^k \cdot 2|\mathcal{P}|$.

Candidates are enumerated shallowest-first, so a truncated answer is a prefix rather than a sample, and truncation is reported: a partial answer must never be readable as a complete one.

Remark 7.3 (Why the domain is restricted). The set $\{\varphi : s \models B_a \varphi\}$ is infinite—conjunction alone generates without bound—so “everything a believes” is not enumerable, and no implementation choice changes this. ML_d is the restriction adopted by Muise et al. (2015) for depth-bounded epistemic planning; here it serves the same purpose, being finite with a size determined by the signature and the depth.

Remark 7.4 (What the restriction costs). The restriction concerns Boolean structure *inside* a modality only; structure outside is untouched, because the pattern is an arbitrary formula. $Kw_a \varphi$ is by definition $K_a \varphi \vee K_a \neg \varphi$, and enumeration reaches it either through the sugar or written out as $K[a] _ \mid K[a] !_$, the two holes taking one filler.

Inside a modality, conjunctive candidates would be redundant, since K and B are normal and $K_a(\varphi \wedge \chi) \equiv K_a \varphi \wedge K_a \chi$. Disjunctive candidates would *not* be: $B_a(p \vee q)$ may hold when

Question	Pattern
What does a believe?	$B[\text{alice}] _$
What can a not settle?	$?[\text{alice}] _$
What does a think c believes?	$B[\text{alice}] \ B[\text{carol}] _$
What does a believe that is false?	$B[\text{alice}] _ \& !_$
Where do a and c disagree?	$B[\text{alice}] _ \& B[\text{carol}] \ !_$
What does a know that c does not?	$K[\text{alice}] _ \& !K[\text{carol}] _$
What does c believe without knowing?	$B[\text{carol}] _ \& !K[\text{carol}] _$

Table 7: Enumeration patterns. The repeated hole is what makes the last four expressible; each occurrence receives the same filler.

neither B_ap nor B_aq does, which is the state of knowing that one of two things holds without knowing which. That is the one genuinely missing class, and the natural direction in which to widen enumeration.

Table 7 illustrates why uniform filling of repeated holes is the design’s load-bearing choice. The false-belief pattern $B_a_ \wedge \neg_$ asks for exactly the conjunction that Section 1 identified as the object of the Sally–Anne task, and it finds instances the analyst had no reason to anticipate.

Example 7.5 (Enumeration on the running example). This is task R6 of Section 1.1, on the scenario that raised it. Ask first what Ana believes that is not so:

```
$ delhi ask search_rescue.delhi -d 1 -q "B[ana] _ & !_"
B[ana] (!trapped) & !(trapped)
B[ana] (K[drone] !trapped) & !(K[drone] !trapped)
B[ana] (B[drone] !trapped) & !(B[drone] !trapped)
3 of 28 candidates at depth 1
```

The first is the stale sweep we put there. The other two were not designed in and are consequences of it: taking the wing to be clear, she also takes the drone to have found it clear. An analyst checking a guessed formula would have found the first and missed the others.

The more useful question is the drone’s own. It is not *what is false* — the drone cannot see that from inside a colleague’s head — but *where does a colleague’s belief contradict what I know*. That is a pattern with two holes and two different operators:

```
$ delhi ask search_rescue.delhi -a "radio_fails()" "drone_reports()" \
-d 1 -q "B[ana] _ & K[drone] !_"
B[ana] (linked) & K[drone] !(linked)
B[ana] (K[ben] linked) & K[drone] !(K[ben] linked)
B[ana] (B[ben] linked) & K[drone] !(B[ben] linked)
B[ana] (K[drone] linked) & K[drone] !(K[drone] linked)
B[ana] (B[drone] linked) & K[drone] !(B[drone] linked)
B[ana] (B[ben] trapped) & K[drone] !(B[ben] trapped)
6 of 28 candidates at depth 1
```

Six divergences, of which the drone had been told about none. Five concern the radio: Ana did not see it fail, so she is wrong about that and about what Ben and the drone make of it. The sixth is the one that matters operationally— $B_{ana}B_{ben} \text{trapped}$ while $B_{ben} \text{trapped}$ is false. Ana is about to act on the assumption that Ben heard the call. This is a false belief about a *mind* rather than about the world; it arose from one observability condition rather than from anything declared; and no one had to guess its shape to find it.

8 Implementation

`delhi` is implemented in Rust in six crates: a hash-consed formula store, a set of shared traits, the semantic core, the front end, the command-line tool, and an optional browser interface. Every crate but the last has no external dependencies.

8.1 Representation

Three decisions account for most of the performance behaviour reported in Section 9.

Hash-consed formulas. Formulas are interned in a store that guarantees structural identity implies pointer identity. Subformula equality is therefore an integer comparison, and memoisation tables can be keyed on formula identifiers.

Bitset models. Valuations are bitsets over $\mathcal{P}$ and relations are bitsets over $W \times W$ per agent. The dominant cost of update is the edge-condition evaluation of Definition 6.5, performed at two worlds per edge per agent; representing the relation densely makes that loop cache-friendly at the cost of $|\mathcal{G}| \cdot |W|^2$ bits.

Memoised entailment. Satisfaction is computed by memoised recursion keyed on (world, formula). Common knowledge is evaluated as a reachability closure rather than by unfolding.

8.2 Bisimulation contraction

Product update multiplies: each action crosses every world with every distinguishable event, so iterated update grows exponentially unless the model is quotiented. `delhi` implements two notions.

Definition 8.1 (Two bisimulations). $\leftrightarrow_R$ is the standard relational bisimulation refining on R_i and R_i^{-1} in both directions. $\leftrightarrow_D$ refines on the *derived* relations $\sim_i$, $\rightarrow_i$ and R_i that the satisfaction clauses of Definition 4.2 actually use.

Proposition 8.2. $\leftrightarrow_R$ is sound—bisimilar states satisfy the same formulas of $\mathcal{L}_G^P$ —and is a congruence for product update. It is not complete: it may distinguish states that no formula separates.

Proof sketch. *Soundness* is the standard induction on formula structure. The atomic case is the valuation clause of the bisimulation. Each modal case needs its quantification domain to be preserved, and each is built from R_i : $\sim_i$ is its symmetric closure, $\rightarrow_i$ is definable from it by maximality, and C_g is the closure of a union of $\sim_i$. A relation respecting R_i in both directions therefore respects all four, and the back-and-forth conditions transfer the induction hypothesis across each.

Congruence. Let Z be a $\leftrightarrow_R$ -bisimulation between M and M' and let A be an action model. The relation $Z' = \{(\langle u, e \rangle, \langle u', e \rangle) : (u, u') \in Z\}$ —pairing product worlds built from the *same* event—is a bisimulation between the updates. Preconditions are preserved because they are formulas of $\mathcal{L}_G^P$ and soundness applies; valuations are preserved because *add* and *del* depend only on e ; and the edge predicate $e \xrightarrow{iuv} f$ is preserved because $Q(e, f)(i) \in \mathcal{L}^P$ and bisimilar worlds agree on propositional formulas. Each disjunct of Definition 6.5 is then a Boolean combination of preserved conditions.

Incompleteness is exhibited rather than argued: Section 8.2 identifies the cause, and the gap is measured in Section 9. $\square$

$ W $	agents	models	incomplete	rate
2	1	8	0	0%
2	2	32	0	0%
3	1	115	6	5.22%
3	2	2,645	144	5.44%
4	1	2,595	264	10.17%
4	2	448,935	42,120	9.38%

Table 8: Incompleteness of $\leftrightarrow_R$, measured by *exhaustive* enumeration of all locally well-preordered models of the given size—not by sampling. “Incomplete” counts models in which $\leftrightarrow_R$ separates at least one pair that modal equivalence identifies.

The incompleteness has an identifiable cause. $\leftrightarrow_R$ refines against R_i^{-1} , whereas no operator of Definition 4.2 quantifies over the inverse relation; the extra refinement is therefore invisible to the language. $\leftrightarrow_D$ removes it and coincides exactly with modal equivalence, but its status as a congruence for update is open, so *delhi* uses $\leftrightarrow_R$ wherever a contracted state is fed back into an update, and offers $\leftrightarrow_D$ for measurement.

Table 8 quantifies the gap the source literature leaves as a remark. Two observations follow from it.

The rate is bounded and small, reaching 10.17% at four worlds. Against this, $\leftrightarrow_R$ and $\leftrightarrow_D$ produce *identical* world counts on every domain in our example corpus and at every cycle of every table in Section 9—so the measured gap, though real, does not appear on structured domains.

The incompleteness is not a multi-agent phenomenon, contrary to how it is usually described. The single-agent rates (5.22%, 10.17%) are as high as the multi-agent ones (5.44%, 9.38%). Since the technique the incompleteness is normally attributed to is complete in the single-agent case, an implementation exhibiting single-agent incompleteness is not running that technique at all, but plain partition refinement over the two directions of R_i . That is a diagnosis of the cause, and it is what identifies R_i^{-1} as the culprit.

Soundness was probed separately: across 451,730 exhaustively enumerated models plus 24 million randomly generated ones ($5 \leq |W| \leq 8$, two to three agents), no pair merged by $\leftrightarrow_R$ was separated by $\leftrightarrow_D$, consistent with Proposition 8.2.

9 Evaluation

We report model growth and elapsed time under iterated update, with and without contraction.

Protocol and its limits. All figures come from *delhi*’s own *bench* subcommand, which applies a named action list repeatedly and records world count and cumulative elapsed time after each cycle. Runs are single-threaded, release-profile builds (`1to = "fat"`, one codegen unit), on one x86-64 laptop. **Each figure is a single run, and we report no variance.** That is adequate for the claims made here, all of which concern growth of the *world count*—an exact, deterministic integer, not a timing—and none of which turns on a timing difference smaller than an order of magnitude. It is not adequate for comparing *delhi* against another system, and we do not attempt such a comparison: Tables 12 and 13 position *mB⁺* against EFP 2.0 (Le et al., 2018) and PDKB (Muisse et al., 2015) on expressiveness and machinery, not on speed. Those systems solve a different problem—plan search—and a model checker’s per-update cost is not comparable to a planner’s time-to-plan.

The world counts below are reproducible exactly; the timings should be read only as orders of magnitude.

Cycle	No contraction		$\leftrightarrow_R$		$\leftrightarrow_D$	
	worlds	cumulative	worlds	cumulative	worlds	cumulative
0	2	—	2	—	2	—
1	16	72.9 μ s	14	205.7 μ s	14	168.7 μ s
2	128	2.2 ms	16	797.7 μ s	16	903.3 μ s
3	1024	133.6 ms	16	1.4 ms	16	1.7 ms
4	8192	9.47 s	16	2.0 ms	16	2.5 ms

Table 9: *Coin Lie*, one cycle being its three actions. Uncontracted growth is multiplicative in the number of distinguishable events; cost grows faster still, because the relation is $|\mathcal{G}| \cdot |W|^2$ bits and almost every operation traverses it.

Cycle	No contraction		$\leftrightarrow_R$		$\leftrightarrow_D$	
	worlds	cumulative	worlds	cumulative	worlds	cumulative
0	8	—	8	—	8	—
1	32	205.8 μ s	24	351.4 μ s	24	372.1 μ s
2	128	2.5 ms	32	1.2 ms	32	1.3 ms
3	512	31.5 ms	32	2.4 ms	32	2.6 ms
4	2048	490.3 ms	32	3.5 ms	32	3.9 ms
5	8192	8.01 s	32	4.7 ms	32	5.2 ms

Table 10: *Muddy Children*, three agents; one cycle is `father_speaks()` then `nobody_knows()`. A second domain reaching a fixed point, at a different size (32 rather than 16), which is what makes the Coin Lie result look like a property of contraction rather than a coincidence of one scenario. Table 11 is the counterexample to that reading. Note that both quotients agree at every cycle, including the transient 24 at cycle 1.

9.1 Growth without contraction

Table 9 shows the uncontracted trajectory reaching 8192 worlds and 9.47 seconds by the fourth cycle—already unusable—while both contracted trajectories settle at 16 worlds and roughly 0.6 ms per cycle, making total cost linear in the number of actions rather than exponential.

Table 10 records *Muddy Children* behaving the same way and settling at a different size, which is the reason to report it: one domain reaching a fixed point is an anecdote, two at different sizes suggest a pattern. Section 9.2 then breaks it.

9.2 A fixed point is not the general case

It is tempting to conclude that contraction bounds model size. It does not.

Table 11 records a domain in which contraction does not reach a fixed point. This is not a deficiency of the quotient: the worlds genuinely differ, and any sound contraction must keep them apart. The practical consequence is that contraction is necessary but not sufficient, and that a forward-search planner built on this semantics will require depth bounding or an additional abstraction, not merely a canonical state key.

9.3 Correctness

The implementation carries 287 tests. These include the frame properties of Theorem 4.4 as property tests over generated models, the identity $B_i^\top \varphi \equiv B_i \varphi$, the divergence of Remark 6.7 as a pinned regression, and, for each example domain, an assertion of the headline claim the

Cycle	No contraction		$\leftrightarrow_R$	
	worlds	cumulative	worlds	cumulative
0	8	—	8	—
1	36	524.9 μ s	36	522.6 μ s
2	160	5.4 ms	92	3.9 ms
3	756	95.5 ms	188	23.8 ms
4	3740	2.36 s	588	119.8 ms
5	6052	5.96 s	2460	1.29 s
6	(cap reached)		7276	9.12 s

Table 11: *Grapevine* under $\leftrightarrow_R$ contraction. The cycle—an agent leaves, a secret is shared, the agent returns—creates a genuinely new distinction each time round, because each repetition adds a layer of who-was-present-for-what. Contraction cannot merge worlds that are really different.

domain exists to demonstrate—so that the documentation and the implementation cannot drift apart silently.

10 Related work

Epistemic planning has two traditions in tension. One begins from *expressiveness*: dynamic epistemic logic (Baltag et al., 1998; van Ditmarsch et al., 2007) will represent essentially anything, at the cost of requiring the modeller to hand-build event models and of undecidable plan existence (Bolander and Andersen, 2011). The other begins from *tractability*, restricting the representation until established planning machinery applies—the $m\mathcal{A}^*$ family of Baral et al. (2022), the forward-search planners of Le et al. (2018), and the depth-bounded compilation to classical planning of Muise et al. (2015). Action languages occupy the middle: they retain a semantics grounded in DEL while deriving event models from declared observability. $m\mathbf{B}^+$ is in this third group, and specifically in the branch that replaced knowledge with belief.

	Belief $\neq$ knowledge	Revision on contradiction	2nd-order false belief	False belief about observers	B_i^ψ / $\Box_i$
DEL	yes	via update rules	yes	yes	in extensions
Baltag & Smets	yes	yes (its origin)	yes	—	yes
$m\mathcal{A}^*$	limited	collapses all uncertainty	no	no	no
$m\mathcal{A}^*$ + h.o. obs.	limited	as $m\mathcal{A}^*$	yes	yes	no
$m\mathbf{B}$	yes	yes, preserving other uncertainty	yes	yes	models only
$m\mathbf{B}^+$ / delhi	yes	yes	yes	yes	yes, in the language
EFP / EFP 2.0	knowledge- oriented	—	—	—	no
PDKB / RP-MEP	yes	bounded	to the depth bound	no	no

Table 12: Expressiveness. “Limited” for $m\mathcal{A}^*$ records a deliberate trade of expressiveness for planning performance, not a shortcoming.

	Event models	State representation	Planner
DEL	hand-built per problem	Kripke models	none; existence undecidable
Baltag & Smets	action-priority update	plausibility models	none (a logic)
$m\mathcal{A}^*$	derived from observability	Kripke models	yes (ASP or forward search)
mB	derived from observability	plausibility models	yes
delhi	derived from observability	plausibility models, bitset-backed	not yet
EFP 2.0	derived	possibilities / Kripke	yes, heavily optimised
PDKB	—	depth-bounded knowledge bases	yes, compiled to classical

Table 13: Machinery. The planner column is where **delhi** is behind: EFP 2.0 and PDKB are mature planning systems, whereas **delhi** is a model checker with the components for a planner present but unused.

10.1 What is transcribed and what is new

Intellectual honesty requires separating the two. Transcribed from Buckingham (2021) and the associated papers: the plausibility-model semantics, the six statement forms of Definition 6.1, the observability model including local dynamic observability, the three constructions of Table 6, and the product update of Definition 6.5. Taken from Baltag and Smets (2006, 2008): the operators $\Box_i$ and B_i^ψ and the action-priority discipline. The name **mB**⁺ is ours and appears in no source.

New here, to the best of our knowledge:

1. Lifting $\Box_i$ and B_i^ψ into the object language of an action formalism, making them queryable properties of a planning state rather than metatheoretic notions (Section 4), together with the introspection analysis of Theorem 4.11 and Corollary 4.12.
2. The enumeration procedure of Definition 7.2: patterns with repeated holes filled uniformly, over the modal-literal domain of Muise et al. (2015) used as a *search space* rather than as a state representation.
3. Declarative initial states (Section 5.2) with post-construction verification, and the auxiliary language layers—definitions, Horn rules over constants, invariants.
4. Conditional ontic effects combined with list-valued **causes**, which the two source presentations offer only separately.
5. The measurement of Section 9, in particular the demonstration that contraction reaches a fixed point on some domains and provably not on others, and the diagnosis of $\leftrightarrow_R$'s incompleteness as refinement against an inverse relation no operator uses.

11 Conclusion

We have given the syntax and semantics of **mB**⁺ and of the **delhi** language that realises it, covering the static logic, the surface language, the action layer, and a query language with an enumeration procedure. Three findings seem to us to have interest beyond the implementation.

First, safe belief behaves quite differently from knowledge and belief with respect to introspection: $K_i\Box_i\varphi \equiv K_i\varphi$ and $B_i\Box_i\varphi \equiv B_i\varphi$, so an agent can never certify a safe belief short of knowledge and unconditionally believes its beliefs to be safe. Safe belief is thus an analyst's operator, not an agent's.

Second, the choice of product-update rule is not a matter of taste. The two rules in the literature diverge precisely on the configuration that the standard demonstration scenario exercises, and one of them prevents that scenario's conclusion from arising at all.

Third, bisimulation contraction is necessary but not sufficient for tractable iterated update. It converts exponential growth into a fixed point on several standard domains and demonstrably does not on others, which bears directly on the design of any planner built over this semantics.

Future work. The obvious next component is a forward-search planner over the contracted state space; the canonical state keys and the contraction it requires are already present. Two open questions bear on it: whether $\Leftrightarrow_D$ is a congruence for product update, which would allow the tighter quotient to be used inside search; and how to bound or abstract domains of the *Grapevine* kind, where sound contraction cannot prevent growth. On the language side, the natural extension is to admit formula-valued postconditions in the style of Andersen et al. (2012), which subsume conditional effects, together with a determination of what observers should learn from them.

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